

ZTF-Orchestrator

User Guide

Initial Configuration & Operation

Version 1.2.4 | John Goulden | 2026-05-21

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1. Overview

ZTF-Orchestrator is a web-based interface for the Nutanix ZeroTouch Framework. It provides a visual operator interface for automating Nutanix infrastructure deployments including cluster creation, node imaging, Prism Central deployment, and workload provisioning without requiring command-line access to the underlying Python framework.

Key capabilities:

- 13 pre-built workflow templates with guided form-based configuration
- 61 individual atomic scripts organised across 12 categories
- Global credential and IPAM configuration (Local / CyberArk / Infoblox)
- Config file management with automatic 5-version backup and restore
- Real-time execution streaming with live terminal output
- Execution history with re-run capability
- Workflow dry-run for pre-flight validation before committing
- Role-based access control: admin, operator, viewer
- Webhook notifications on workflow completion

2. Signing In

Open a browser and navigate to <http://localhost:5001> (or the configured port). On first launch the server generates a random admin password and prints it to the terminal.

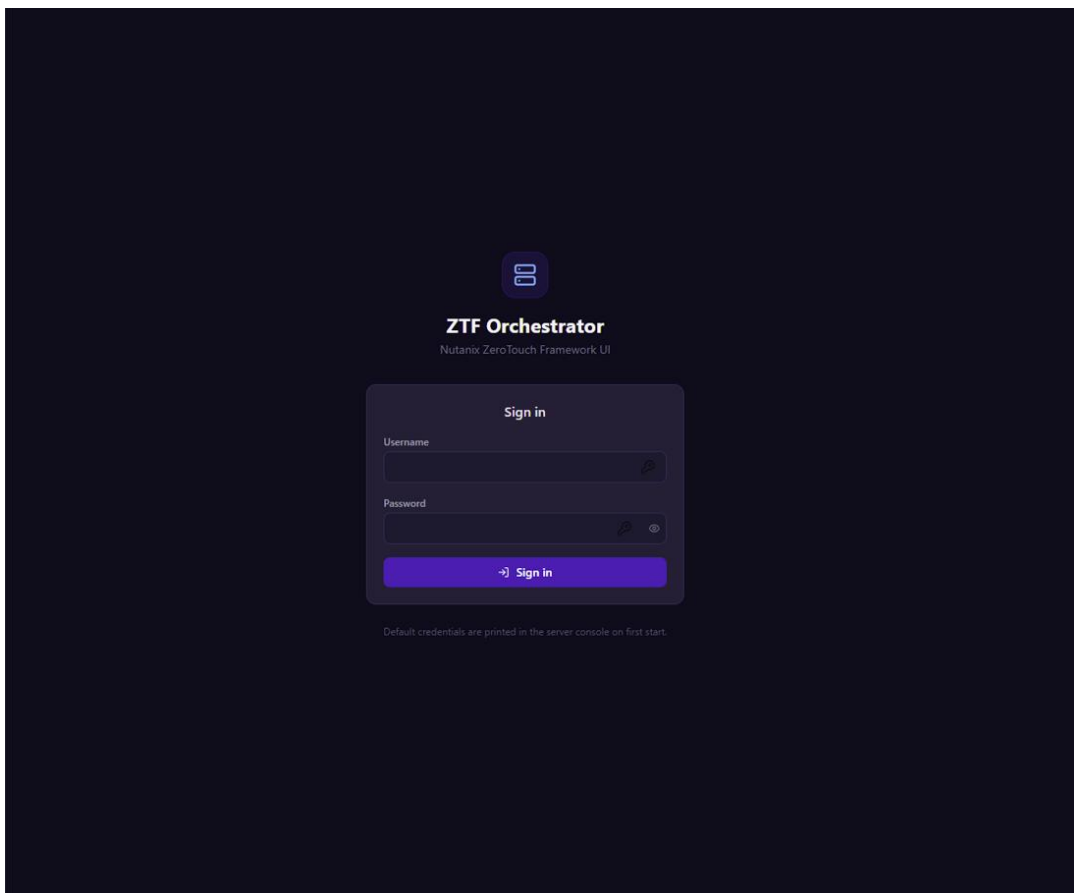


Figure: Login screen

Enter the username admin and the password printed in the server console, then click Sign in.

First Run: Copy the password from the terminal before navigating to the browser. If missed, delete users.json from ~/.ztf-ui/ and restart the server to regenerate credentials.

3. Dashboard

The Dashboard is the home screen. It shows the system health status, recent execution history, execution statistics, and quick-action buttons for the most common workflows.

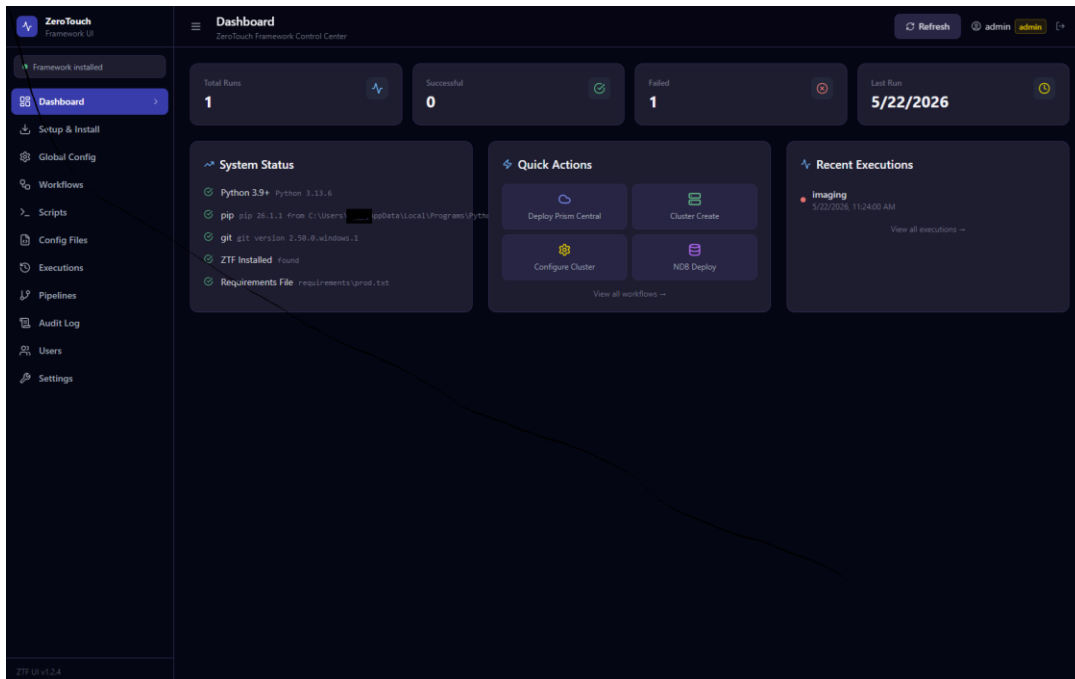


Figure: Dashboard - system status, stats, and quick actions

System Status

The System Status panel confirms that all prerequisites are met:

- Python 3.9+: the Python interpreter on PATH
- Pip: package manager
- Git: required for the Setup & Install wizard
- ZTF Installed: confirms main.py is found at the configured ZTF path
- Requirements File: confirms the ZTF dependency file exists

Note: The green Framework installed badge in the top-left sidebar indicates ZTF is ready to use. If it shows amber, navigate to Setup & Install to install the framework first.

Auto-refresh

The dashboard refreshes automatically every 30 seconds. Use the Refresh button in the top-right corner to trigger an immediate update.

4. Setup & Install

Use the Setup & Install page to install or reinstall the ZeroTouch Framework. This page is only required once during initial deployment, or when updating the framework.

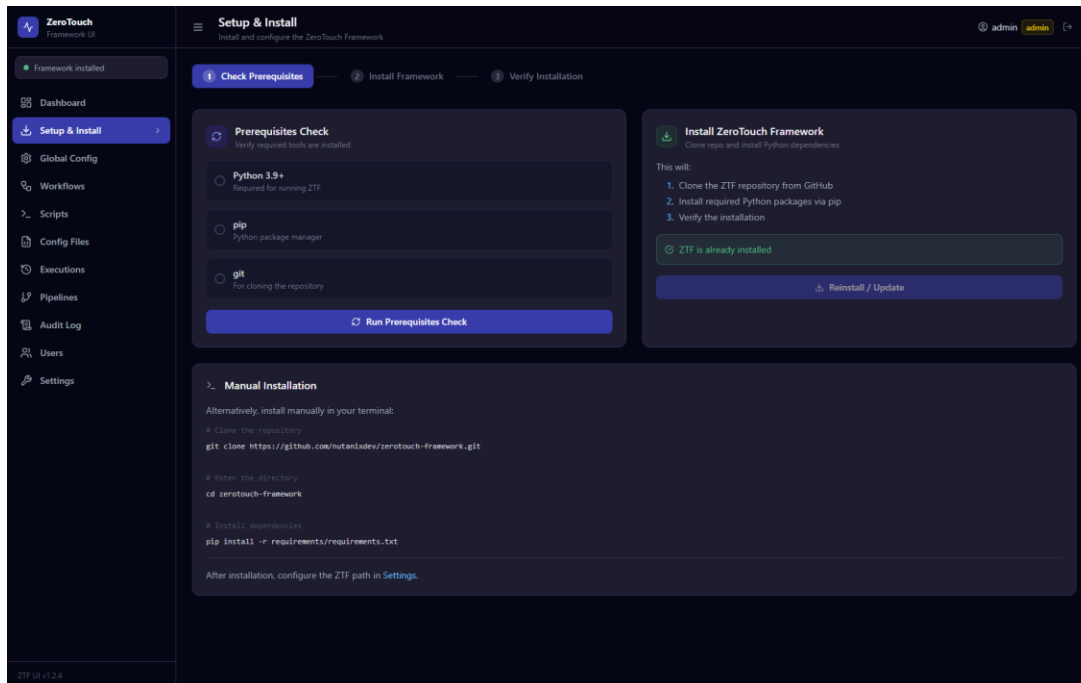


Figure: Setup & Install - prerequisites check and installation

Prerequisites Check

Click Run Prerequisites Check to verify Python, pip, and git are available. Each item is confirmed with a green tick before installation proceeds.

Install Framework

Click Install Framework to clone the ZTF repository from GitHub and install its Python dependencies. Progress streams live in the terminal panel. For air-gapped environments, see the Field Installation Guide.

5. Global Configuration

Global Config defines the credentials, vault type, and IP allocation method used across all workflows. Changes are saved to global.yml in the ZTF framework directory.

Credentials Tab

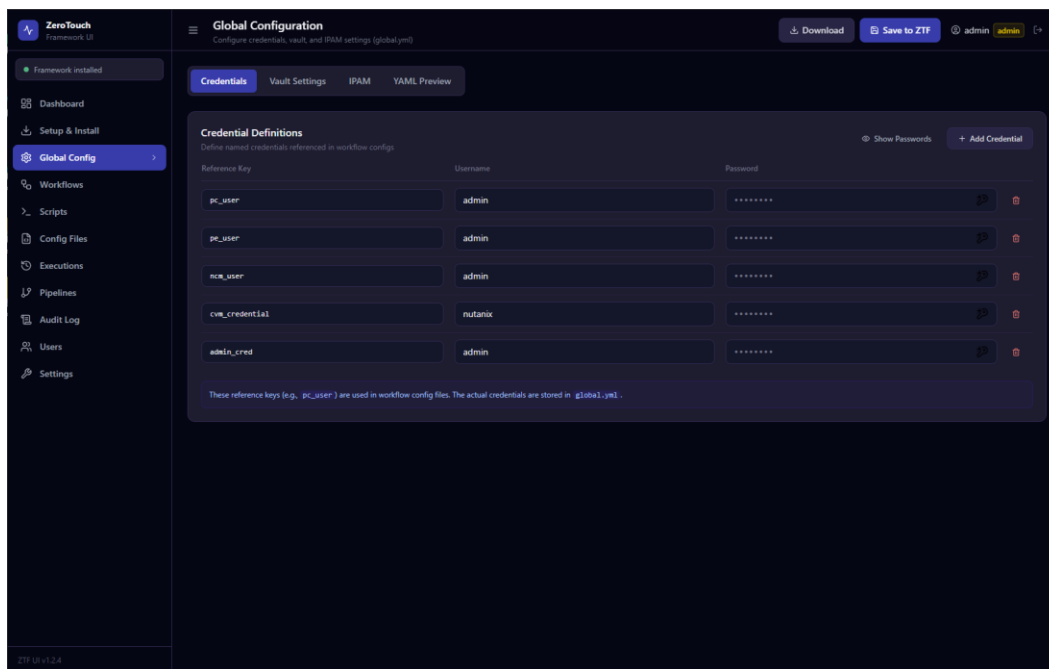


Figure: Global Config Credentials tab

Each row defines a named credential reference. The reference key (e.g. pc_user, cvm_credential) is used in workflow config files to look up the actual username and password stored here. Click Add Credential to add additional entries.

- Reference Key: the identifier used in workflow YAML files
- Username: the account username
- Password: stored securely in global.yml (masked by default)

Vault Settings Tab

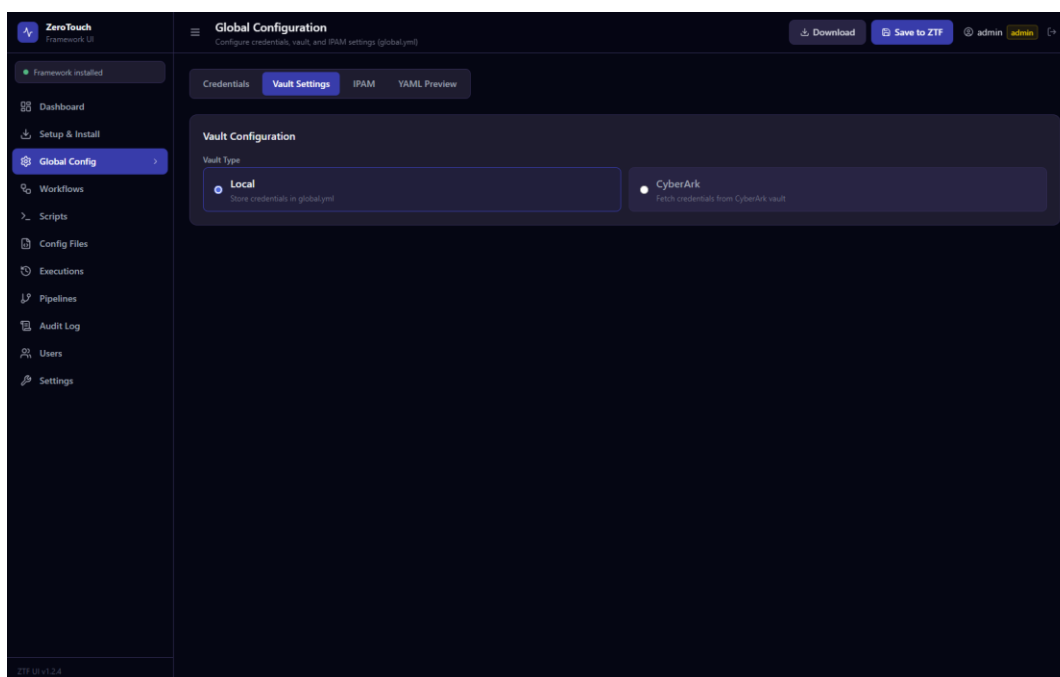


Figure: Global Config - Vault Settings tab

Select the vault type for credential storage:

- Local: credentials stored directly in global.yml (default)
- CyberArk: credentials fetched at runtime from CyberArk; provide host, certificate, and key paths

Note: The IPAM tab configures IP allocation: Static (manually specified per workflow) or Infoblox (automatic allocation). Click Save to ZTF to write global.yml to the framework directory.

6. Workflows

The Workflows page lists all 13 pre-built automation templates, grouped by category. Use the search bar or category filters to find a specific workflow.

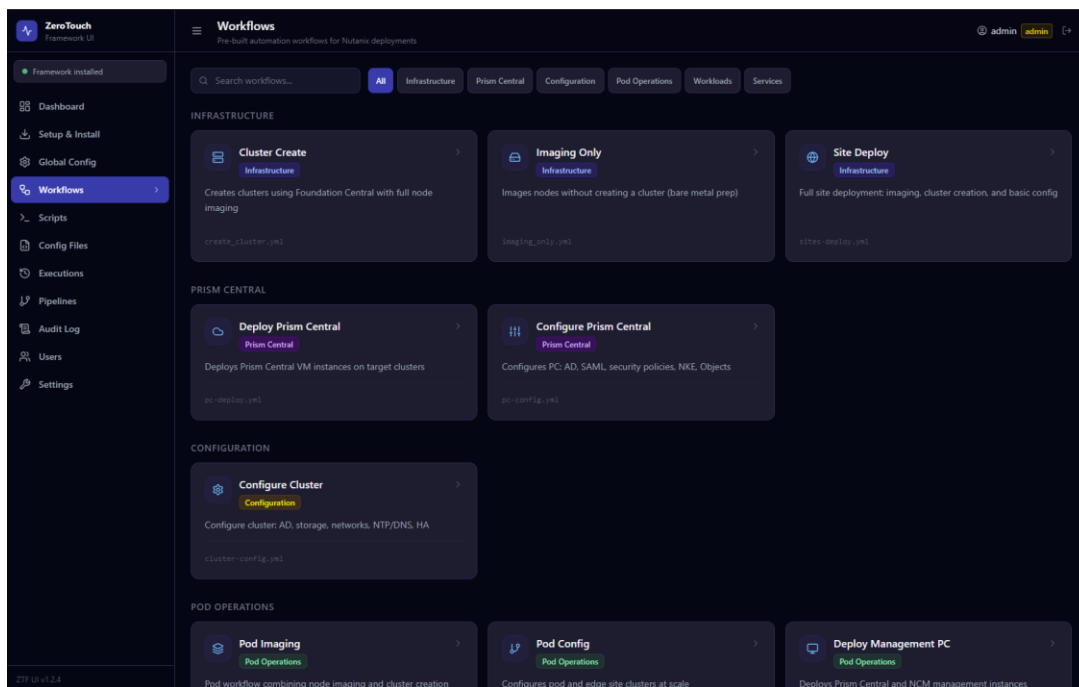


Figure: Workflows list - all 13 templates grouped by category

Category	Workflows
Infrastructure	Cluster Create, Imaging Only, Site Deploy
Prism Central	Deploy Prism Central, Configure Prism Central
Configuration	Configure Cluster
Pod Operations	Pod Imaging, Pod Config, Deploy Management PC, Configure Management PC
Workloads	Calm VM Workloads, Edge AI Workload
Services	NDB Deploy

7. Running a Workflow

Click any workflow card to open the workflow detail page.

Figure: Workflow detail - Cluster Create form with Dry Run and Run Workflow buttons

Configure Tab

Fill in the required fields in the Configure tab. Each workflow has a dedicated form tailored to its parameters. Required fields are marked with an asterisk. Dropdown fields reference the credential keys defined in Global Config.

YAML Preview Tab

Click YAML Preview to inspect the generated configuration before executing. Use Download Config to save the YAML file for offline use or manual execution.

Dry Run

Click Dry Run to run a pre-flight check without executing the workflow. The dry run validates:

- YAML structure: the config is valid and parseable
- Required fields: all mandatory workflow parameters are present
- IP address format: key IP fields contain valid addresses
- TCP reachability: key hosts (Foundation Central, Prism Central, etc.) are reachable on port 9440

Note: Dry Run fires pre-flight checks only. No changes are made to the Nutanix environment.

Run Workflow

Click Run Workflow to execute. A terminal modal opens showing live output streamed from the ZTF framework. The workflow runs until completion; the terminal remains open to review output.

8. Script Library

61 atomic scripts are available for targeted operations adding AD servers, creating subnets, deploying NKE clusters, and more. Scripts can be filtered by category or searched by name.

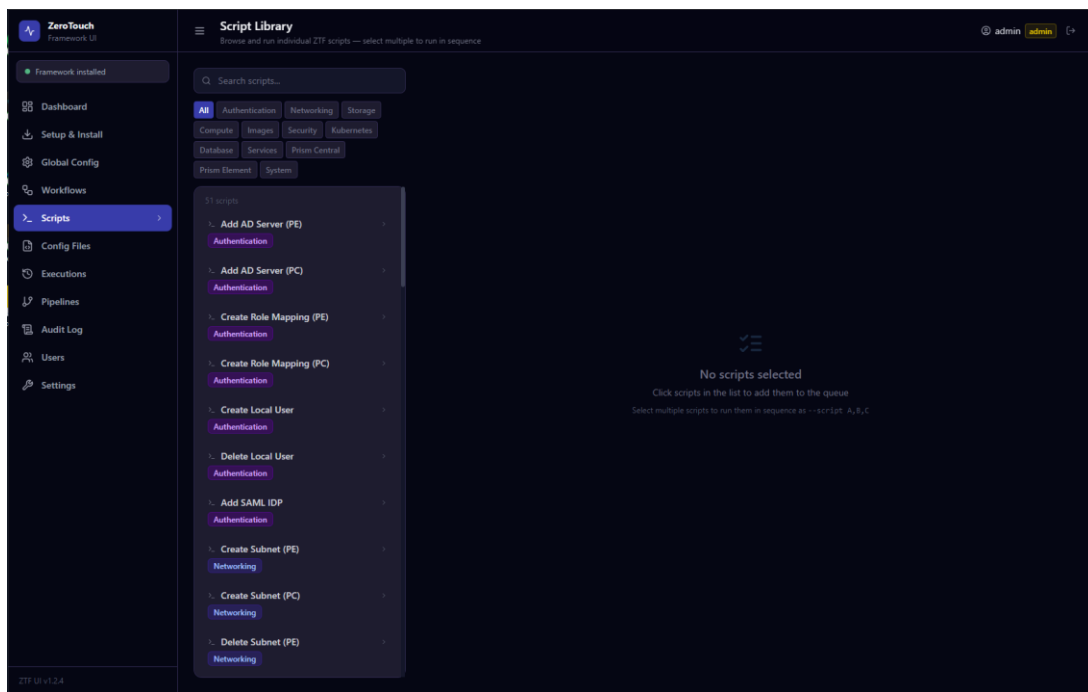


Figure: Script Library - 61 atomic scripts across 12 categories

Select a script from the list, provide the required YAML configuration in the editor panel, then click Run Script. Output streams live in the terminal.

9. Config Files

The Config Files page manages YAML and JSON configuration files used by workflows and scripts. Files are stored in the configured config directory.

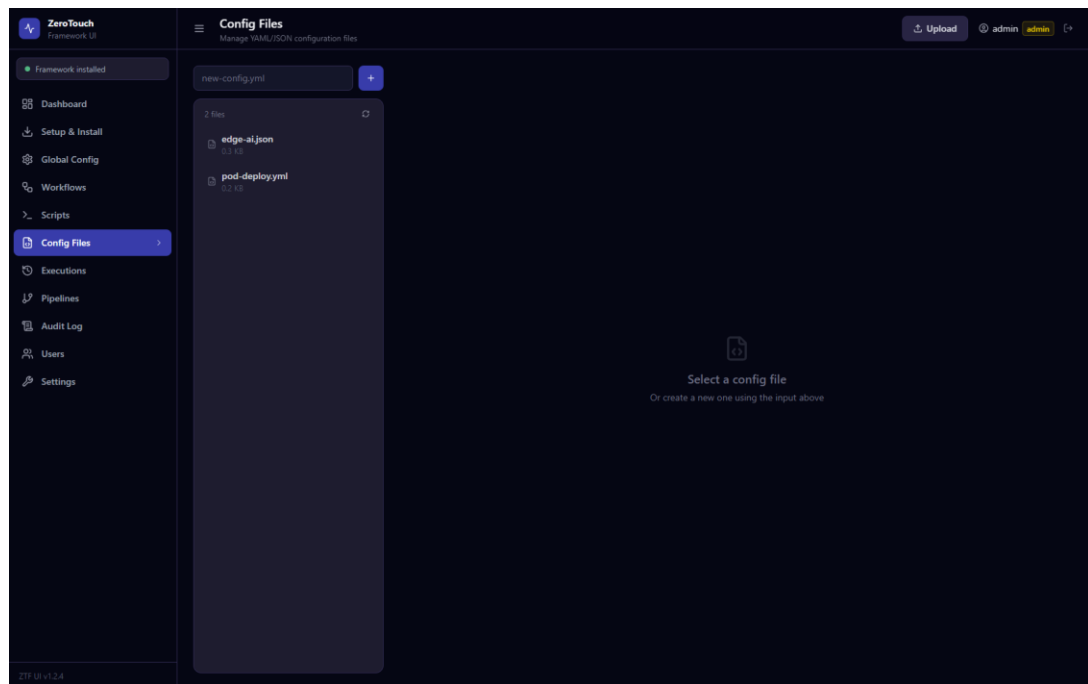


Figure: Config Files - file list with editor and version history

Creating and Editing Files

- Type a filename in the input at the top-left and click + to create a new file
- Click any existing file to open it in the editor
- Edit the YAML/JSON content directly in the editor pane
- Click Save to write the file; a .bak.1 backup is created automatically
- Use Upload to import an existing config file from your local machine

Version History

Each file retains up to 5 previous versions automatically. If backups exist, a History button appears in the action bar showing version count. Click it to view all versions with timestamps and sizes.

- Click Restore on any version to revert the file to that state
- The current file is backed up before any restore operation

Note: Config files are the input passed to workflows at execution time. A workflow reads its config file at the path shown in the YAML Preview tab.

10. Execution History

Every workflow and script run is recorded in Execution History with status, timestamp, and duration. Up to 1000 records are retained.

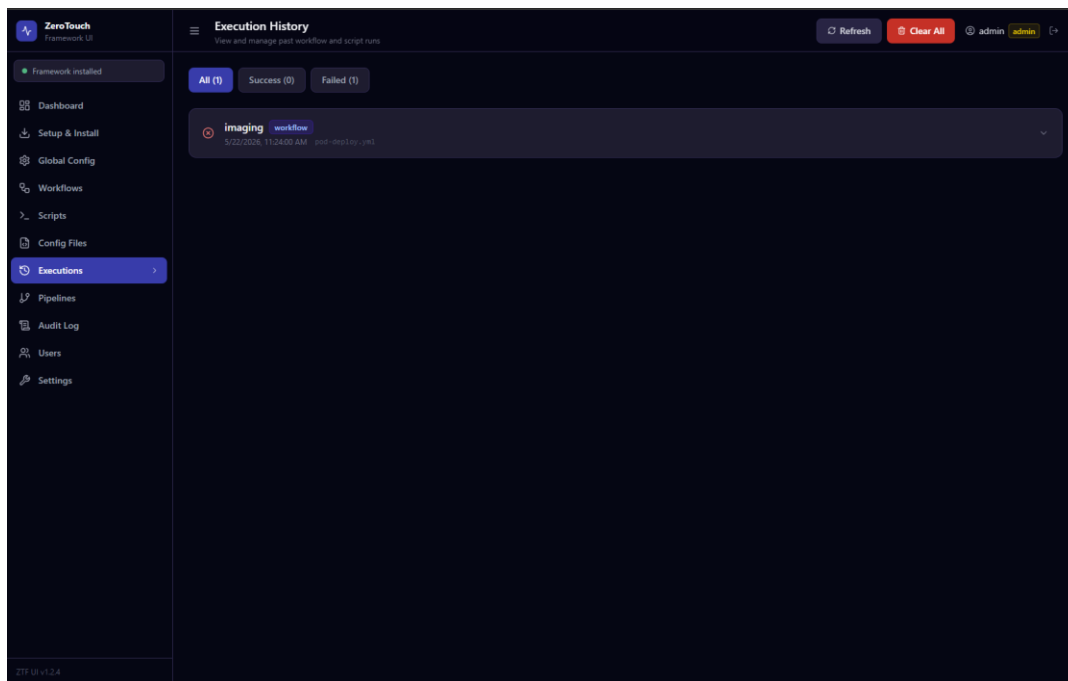


Figure: Execution History - status filters and expandable rows

Filtering

Use the All / Success / Failed filter buttons to narrow the list. Each button shows the count for that status.

Re-run

Expand any execution row by clicking it. If the original config is available, a Re-run button appears alongside the config filename. Clicking Re-run opens the execution modal pre-loaded with the original YAML no form re-entry required.

Note: Re-run is only available for executions recorded after v1.2.2. Earlier records do not store config content.

11. Pipelines

The Pipelines page allows workflows to be chained sequentially with automatic pass/fail gates. Each pipeline runs its steps in order if any step fails, remaining steps are skipped and the pipeline halts. This enables fully unattended end-to-end site deployments without operator intervention between steps.

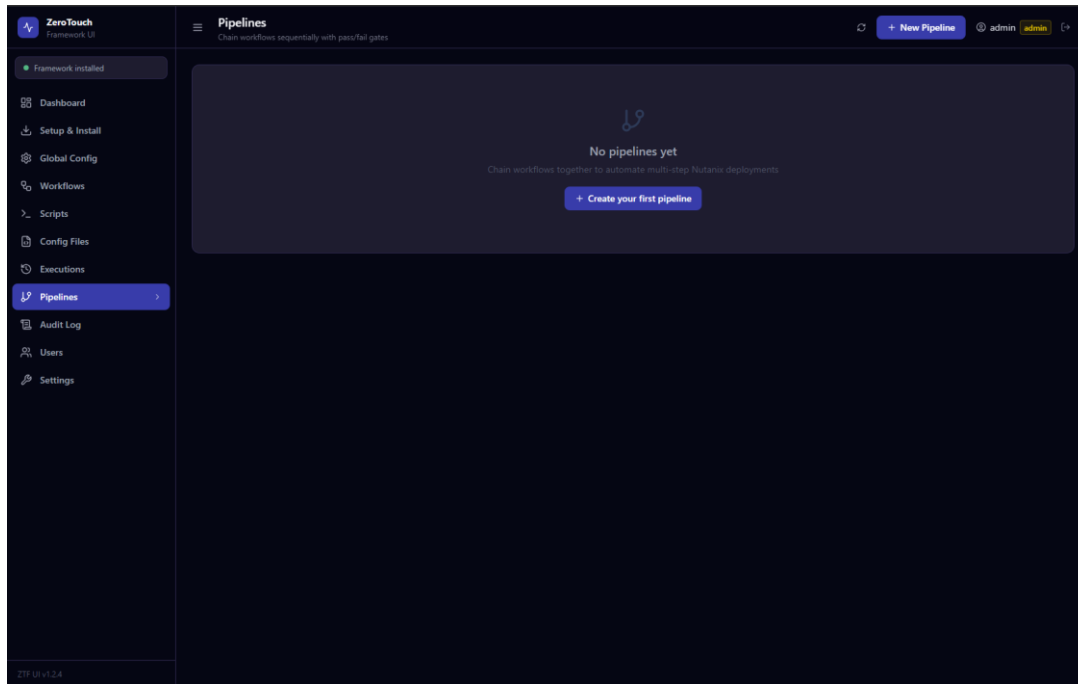


Figure: Pipelines - Chain workflows sequentially with pass/fail gates

Creating a Pipeline

Click New Pipeline to open the builder form.

- Enter a descriptive Pipeline Name (e.g. "Full Site Deployment Site A")
- Click Add Step to append a step to the sequence
- For each step, select the workflow from the dropdown and optionally select the config file to use
- Use the up/down arrows to reorder steps
- Click Create Pipeline to save

Note: Config files listed in the step picker come from the Config Files page. Create and save your workflow configs there before building a pipeline.

Pipeline List

Saved pipelines are displayed as cards showing the pipeline name, step count, last-updated date, and a visual flow of steps connected by arrows (e.g. imaging-only → cluster-create → deploy-pc → config-pc). Each card has Run, Edit, and Delete actions.

Running a Pipeline

Click Run on any pipeline card. The pipeline execution modal opens showing:

- A step progress rail at the top each step shows its status: pending, running (spinner), success (teal), failed (red), or skipped (grey)
- A combined live terminal with step-separator headers between each workflow output
- The modal remains open until the pipeline completes or fails

Pass/fail gate behaviour:

- Every step only starts if the previous step exited with return code 0
- On first failure the pipeline halts; all remaining steps are marked Skipped
- The overall status (success / failed) is recorded in Execution History

Note: Pipeline runs appear in Execution History with a [Pipeline] prefix. Webhook notifications include step-level results if a webhook URL is configured in Settings.

Typical Deployment Pipeline

Step	Workflow	Purpose
1	Imaging Only	Image nodes with AOS and AHV via Foundation Central
2	Cluster Create	Form the cluster and set DNS/NTP
3	Deploy Prism Central	Deploy PC VM to the new cluster
4	Configure Prism Central	Apply AD, SAML, NKE, security policies
5	Configure Cluster	Set containers, networks, and HA reservation

12. Audit Log

The Audit Log page provides a searchable, filterable view of the structured application log. Every significant event logins, workflow executions, config saves, user creation, and settings changes is recorded with a timestamp, the acting user, and their IP address. Access is restricted to the admin role.

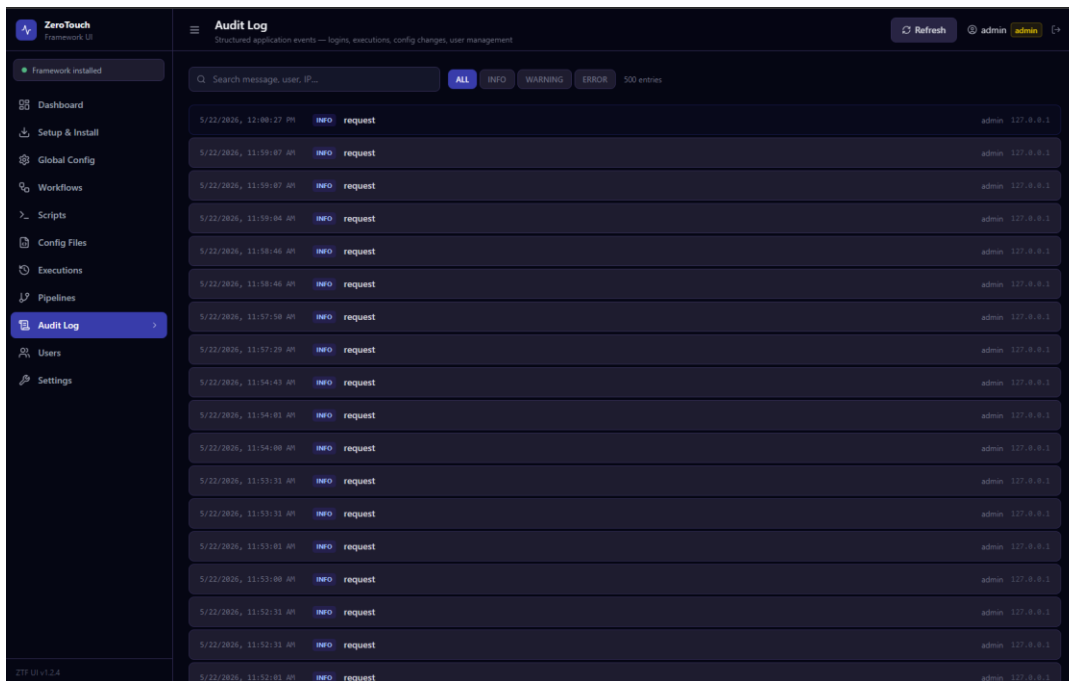


Figure: Audit Log - Structured application events logins, executions, config changes, user management

Note: Audit Log is admin-only. Operators and viewers receive an access-denied response if they navigate to this page.

Reading the Log

Each row shows:

- Timestamp: when the event occurred (local time)
- Level badge: INFO (blue), WARNING (amber), or ERROR (red)
- Message: the event type (e.g. login_success, execution_complete, config_saved)
- User: the account that performed the action
- IP: the client IP address
- Status: success or failed where applicable

Click any row to expand it and see all additional structured fields for that event.

Filtering

- Use the search box to filter by message text, username, IP address, or action name
- Use the level buttons (ALL / INFO / WARNING / ERROR) to narrow by severity
- Click Refresh to reload the most recent entries from disk

Note: The log file is located at ZTF_DATA_DIR/ztf-orchestrator.log (default: ~/.ztf-ui/ztf-orchestrator.log on Linux/macOS, C:\Users\<user>\ztf-ui\ztf-orchestrator.log on Windows). The UI shows the most recent 500 entries by default.

Common Event Messages

Message	Triggered by
login_success	Successful user login

Message	Triggered by
login_failed	Failed login attempt
execution_start	Workflow or script execution begins
execution_complete	Workflow or script finishes (includes status)
pipeline_complete	Pipeline finishes (includes step results)
execution_cancelled	Client disconnected during execution
webhook_fired	Webhook notification sent successfully
webhook_failed	Webhook notification failed

13. User Management

Navigate to Users in the sidebar to manage user accounts. Admin role is required to create, modify, or delete users.

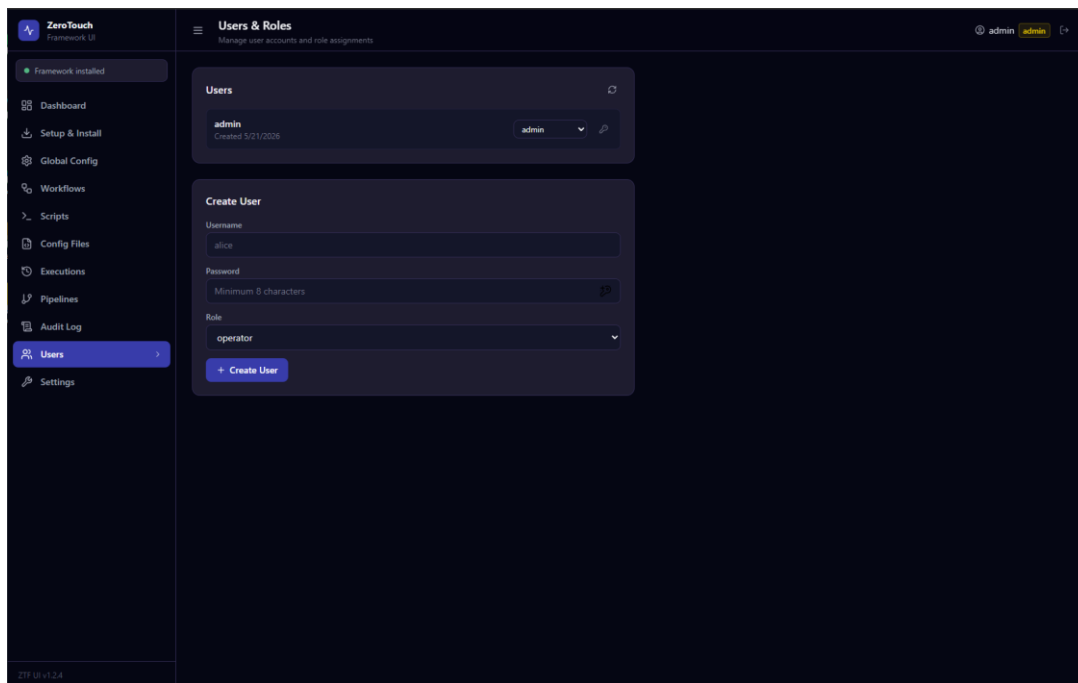


Figure: Users & Roles - user list and Create User form

Roles

Role	Permissions
admin	Full access: settings, global config, user management, all workflows and scripts
operator	Execute workflows and scripts, manage config files, view execution history
viewer	Read-only: view configs, execution history, and system status

Creating a User

- Enter a username and password (minimum 8 characters)
- Select a role from the dropdown

- Click Create User

Changing a Role

Use the role dropdown on any existing user row. The change saves immediately.

Resetting a Password

Click the key icon on any user row to expand the password reset form. Type the new password and press Save or Enter.

Note: Users cannot delete their own account. Admins can delete any other user.

14. Settings

Settings configure the ZTF framework path, Python executable, and notification options. Admin role required to modify.

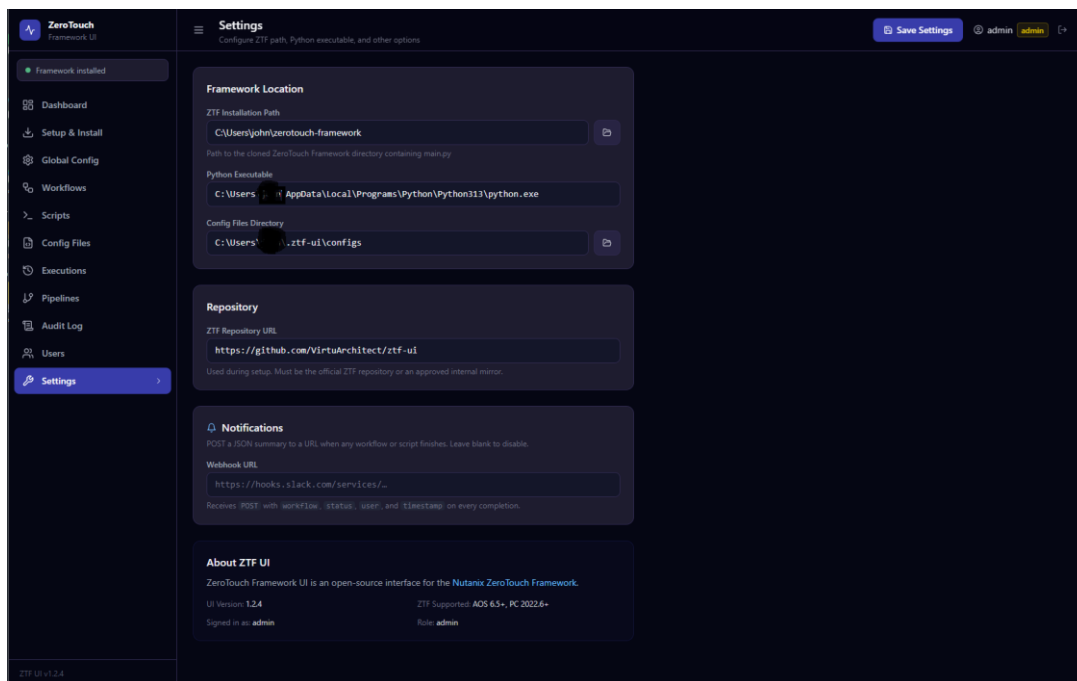


Figure: Settings - framework location, repository, notifications, and about

Framework Location

- ZTF Installation Path: full path to the cloned zerotouch-framework directory containing main.py
- Python Executable: path to the Python interpreter used to run ZTF (defaults to python3)
- Config Files Directory: where YAML/JSON config files are stored (defaults to ~/.ztf-ui/configs)

Repository

The ZTF Repository URL is used by the Setup & Install wizard. For air-gapped deployments, set this to your internal Git mirror URL.

Notifications (Webhook)

Enter a webhook URL to receive a POST notification on every workflow or script completion. The payload includes workflow name, status, user, return code, and timestamp. Compatible with Slack incoming webhooks, Microsoft Teams, n8n, and any custom HTTP endpoint. Leave blank to disable.

Note: Click Save Settings to apply changes. Settings take effect immediately without restarting the server.